CLASS 9 · CBSE · GANITA MANJARI PART 2 · DPP SHEET

Chapter 12: Quadrilaterals

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In a parallelogram $ABCD$, $\angle A = 65^\circ$. Find the other three angles.

2. Two opposite sides of a parallelogram are $(4x + 3)$ cm and $(6x - 7)$ cm. Find x and the length of the side.

3. The diagonals of a parallelogram $ABCD$ meet at E. If $AC = 18$ cm and $BD = 12$ cm, find AE and DE.

4. The angles of a quadrilateral, in order, are 70°, 110°, 70° and 110°. Is it a parallelogram?

5. The sides of a triangle are 8 cm, 12 cm and 16 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.

6. D and E are the mid-points of AB and AC in $\triangle ABC$, and $BC = 15$ cm. Find DE.

7. A median of a triangle is 15 cm long. Into what lengths does the centroid divide it?

8. The diagonals of a quadrilateral are 10 cm and 24 cm. Find the perimeter of its Varignon parallelogram.

9. The diagonals of a quadrilateral are equal. What special kind of parallelogram is its Varignon parallelogram?

10. Can a regular octagon tile the plane on its own? Each of its angles is 135°.

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Answer Key

1. $\angle B = 115^\circ$, $\angle C = 65^\circ$, $\angle D = 115^\circ$
2. $x = 5$; 23 cm
3. 9 cm and 6 cm
4. Yes. Both pairs of opposite angles are equal.
5. $4 + 6 + 8 = 18$ cm
6. 7.5 cm (and $DE \parallel BC$)
7. 10 cm (from the vertex) and 5 cm
8. $10 + 24 = 34$ cm
9. A rhombus
10. No. 360 is not a multiple of 135.

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